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Precision farming for sustainability: An agricultural intelligence model

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ABSTRACT

Digital cultivation is emerging as one of the most promising fields that helped in creating an ecosystem for smart farming. Precision farming, modernized techniques, and creating smart agriculture supply chains are the need of the hour for high-quality yield. Artificial Intelligence (AI) helps create a framework and plays an important role in making decisions by analysing various data points. There are countries where more than 70% of the population depends on agriculture for their living, technological advancements help to improve crop yields and get better farming results through sustainable ways. Each stage in agriculture, starting from preparation of land, crop selection, type of fertilizer to use, to the kind of watering needed for the crops; can be monitored and regulated by technological advancement. Farmers can also make decisions and implement the best practices in their field by using AI and allied technologies. Disruptive technologies such as blockchain, the Internet of Things, remote sensing, imaging technologies, and drones can transform the primitive way of agriculture. Market analysis and user demands can also be foreseen, which helps farmers to get better yields. Another important sector where technology can play a big role in disease control and pest management. Artificial Intelligence-based farming creates high productivity and better yield, increasing individual farmers' profit. In this study, the authors would like to throw light on AI and allied technologies, which can make agricultural productivity increase significantly. In a post-pandemic situation, high-yield and more productive farming will have a major impact. An agricultural intelligence framework model for self-sustained farming is proposed in this work. The proposed framework will help achieve self-sustained growth with increased economic stability. An end-to-end supply chain ensures customers are provided with quality products and farmers are not financially looted. Technology-driven farming will also push the next generation to take up agricultural jobs. The various advancements and strategies we propose in this study aim to build a better ecosystem for transforming Artificial Intelligence into agricultural intelligence.

1. Introduction

Agriculture is the backbone of the economy in nearly all developing countries, with approximately 70%–75% of their populations directly or indirectly dependent on it for their livelihoods (Chandra, 2023). Despite its critical role, it is disheartening to observe that the methods and markets of agriculture often lag behind the necessary standards. Farmers lack proper guidelines and scientific farming methods, leaving them unaware of the latest advancements and technologies. Even agricultural research institutes seem to overlook the development of technologies that could benefit farmers. In stark contrast, Israel is an exemplary model of advanced farming methods that yield high productivity (Albaaji and Vinod Chandra, 2023). In this discussion, we will explore various technological advancements in farming and outline a comprehensive blueprint for establishing an end-to-end smart ecosystem to maximize farm productivity. The role of Artificial Intelligence and expert systems in contributing to a nation's agricultural development cannot be underestimated. As a field of study, Artificial Intelligence involves simulating human intelligence in machines, enabling them to think and act like humans (León-Bravo et al., 2019). This simulation encompasses the capacity to reason and learn to navigate various scenarios successfully.

The challenge is to extend the boundless potential of AI and allied technologies to the ordinary person. Even though the intricacies of artificial intelligence may be beyond the comprehension of the average individual, its impact on society is profound and pervasive, playing an indispensable role in modern life. For instance, imagine an automated system examining a tomato seedling's yellow flower and

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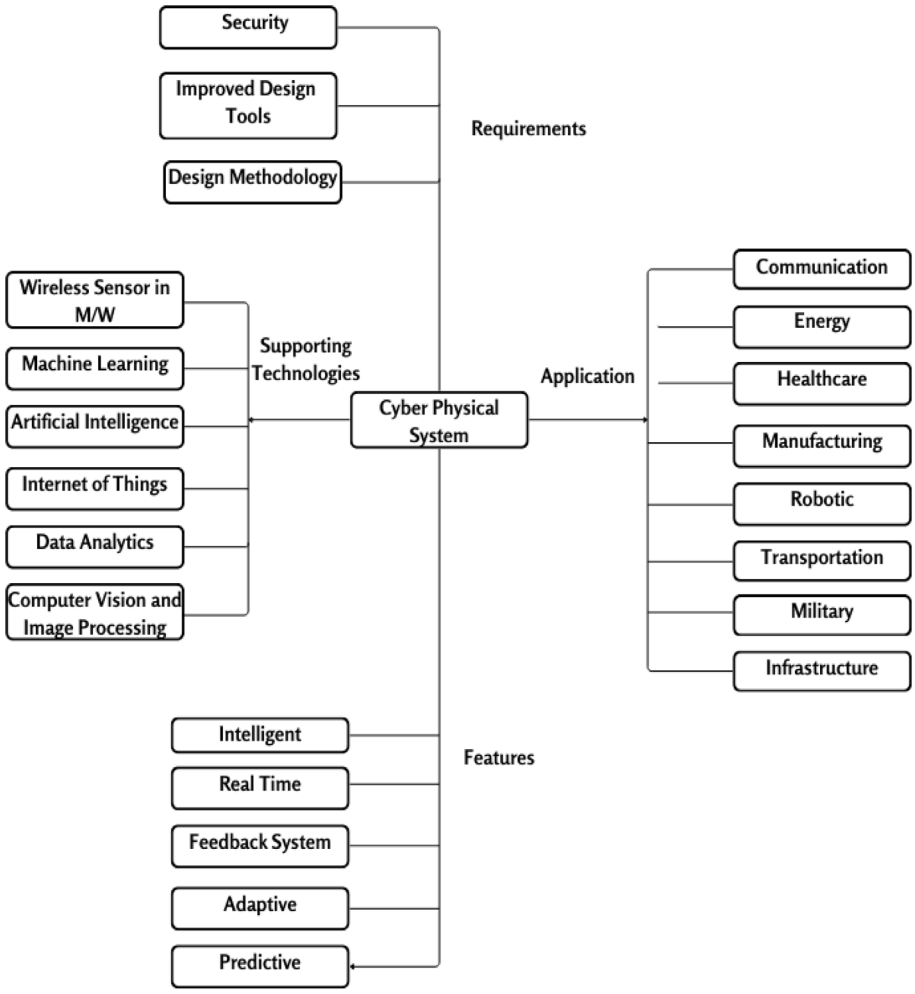


Fig. 1. Cyber physical system framework.

accurately predicting the duration until it ripens into a fruit ready for harvesting, packaging, and distribution to grocery stores (Sharafudeen and Vinod Chandra, 2023). This is not a future vision; it is today's reality. As we gaze into the next decade, we must focus on high-tech agricultural tools, intelligent transportation systems, consistent field analytics for farmers, and innovative pest control mechanisms. AI-powered advancements increase productivity, alleviate farmers' debt burdens through reduced production costs and losses, and minimize the overall environmental impact (Ernst, 2022). Infusing science and technology into agriculture is pivotal for combating severe poverty and famine. Microsoft Corporation, in collaboration with the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT), has embarked on a groundbreaking initiative in Andhra Pradesh, India, since 2018, providing services and solutions to 175 farmers (Albaaji and Chandra, 2024). They developed an Artificial Intelligence sewing application that leverages machine learning and business intelligence from the Microsoft Cortana Intelligence Suite, utilizing cloud-based predictive analytics to empower farmers with essential information and insights. This comprehensive system covers land preparation, seed sowing, climate adaptation, fertilizer application, and the use of organic nutrients, among other aspects. The result has been a remarkable 30% increase in average yield per hectare compared to previous harvests. Predictive analytics are instrumental in assisting farmers in making informed decisions regarding crop sowing, fertilizer application timing, and harvest timing (Ahmad and Sharma, 2023). This involves analysing climate data, historical conditions, market conditions, and individual farming information (Rodríguez et al., 2020). However, it is worth noting that Artificial Intelligence must be entered into the agricultural sector meticulously. Challenges such as unreliable market access, commodity price fluctuations, low incomes, and high debts have made agriculture high-risk. As we reflect on the historical transition from the Industrial Revolution, which reduced the need for human labour, to the present era, it is evident that it is time to introduce intelligent machines (Mor et al., 2021).

Summing up these insights, the central objective of technological advancements should be to establish a cyber–physical framework that allows farmers to maximize their profits by efficiently harnessing their available resources. Cyber-Physical Systems (CPS) represent a novel engineered system class that seamlessly integrates computation and physical processes in dynamic environments. CPS encompasses various technologies, including cybernetics, mechatronics, design and embedded systems, the Internet of Things (IoT), big data, Artificial Intelligence (AI), and more. These systems are intelligent, autonomous, and efficient, and they are poised to drive innovation across various sectors, including agriculture, irrigation, energy, transportation, infrastructure, security, health, and manufacturing (Jararweh et al., 2023).

The distinguishing characteristics of a cyber–physical system include cyber capabilities within every physical component, networking at multiple scales, dynamic reconfiguration, logical and physical coupling of constituent elements, and high levels of automation with control loops operating at various scales. A comprehensive framework that outlines the supporting technologies, requirements, features, and high-level applications is presented in Fig. 1. The expanded framework

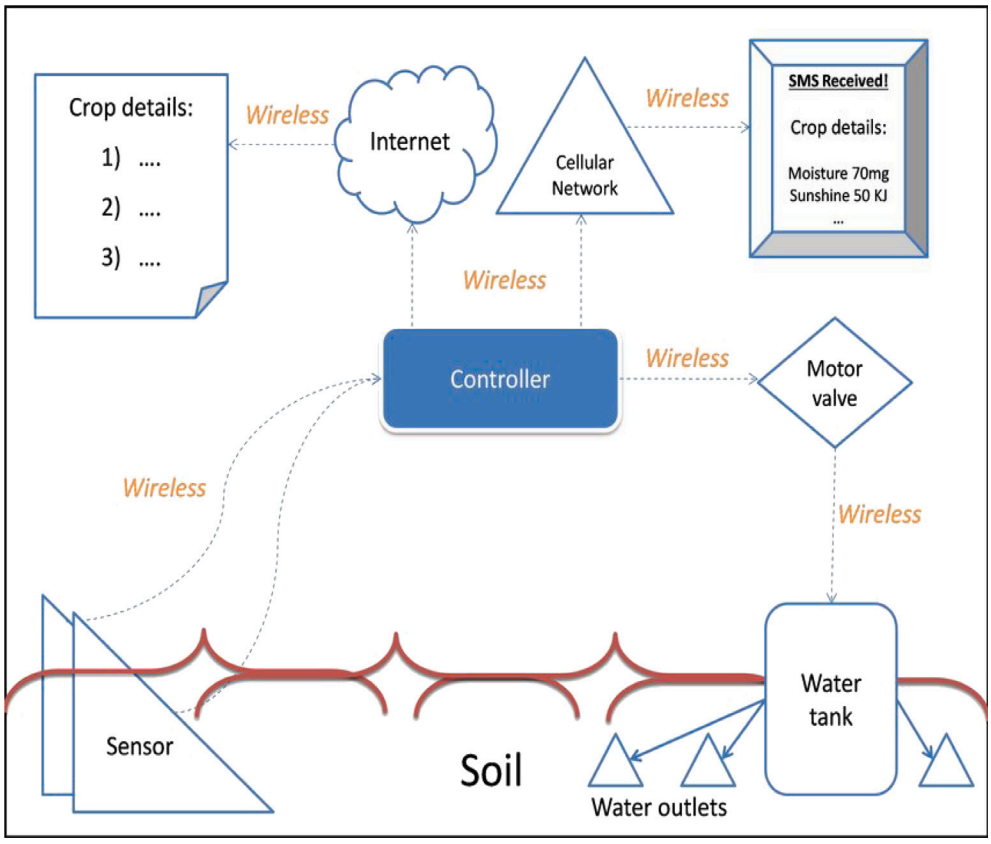


Fig. 2. IoT sensor model.

aims to foster a network where consumers and producers coexist, facilitating the discovery of demand and supply. It is important to emphasize that the proposed framework goes beyond the impact of AI in agriculture; it indirectly promotes a holistic approach to agro-ecology, agro-forestry, and climate conservation agriculture. This holistic approach underscores the pivotal role of technology in ensuring food security, poverty reduction, and environmental sustainability (Talaviya et al., 2020).

2. Technologies in agriculture

Built-in sensors and new technologies that use data have become very important in making many areas of agriculture more efficient. These include checking the quality of seeds, keeping an eye on the soil, predicting the weather, analysing crops, tracking market changes, and setting up distribution networks (Tewari et al., 2020). Integrating cloud computing infrastructure, robust data ecosystems, the Internet of Things (IoT), and machine learning is a promising avenue towards enhancing farming productivity. These technologies are intricately focused on optimizing various facets of agricultural practices, ranging from resource-efficient farming methods, precise irrigation techniques, intelligent chemical application, and streamlined harvesting methodologies to effective pest control strategies. Notably, the advent of Artificial Intelligence-based, AI-based automated irrigation systems has showcased a remarkable surge in the efficacy of the contemporary agricultural supply chain. The need for food grains and the demand for freshwater resources have increased, underscoring AI's crucial role in facilitating intelligent irrigation, monitoring water levels, soil temperature, nutrient content, and predictive weather forecasting (Abioye et al., 2022).

Models must be meticulously crafted to meet future agricultural demands to fulfil these requisites (Sanga et al., 2020). The United Nations Food and Agriculture Organization (UNFAO) has stipulated that by 2050, agricultural productivity must surge by at least 70% to cater to the escalating global population demands. The pressing concerns of natural resource depletion and diminishing arable land have compellingly mandated the enhancement of farm yields. The Internet of Things (IoT) is the networking of physical objects equipped with sensors, processors, software, and other technologies that allow them to connect and share data over the Internet or similar communication networks. In the agricultural landscape, IoT integration necessitates the deployment of diverse sensors for data collection, subsequently archived within sensor-attached memory units for subsequent analysis. However, this paradigm proves less efficacious when confronted with live data mandating real-time responses.

The intrinsic limitations of memory capacity within individual sensors often lead to data inundation, thus advocating a paradigm shift towards cloud-based analytics. Cloud-centric analytics entail transmitting data gleaned from diverse sensors—such as those measuring temperature, soil moisture, and mineral levels—to dedicated cloud-based servers for comprehensive analytics. The resultant analytical outcomes are disseminated through customized local websites or mobile interfaces. For instance, leveraging data obtained from soil moisture and temperature sensors, an automated drip irrigation system can be meticulously devised, ensuring optimal crop hydration regardless of prevailing climatic conditions. As shown in Fig. 2, implementation methods often use technologies like Arduino or microcontrollers. This is shown in a schematic layout that shows an example of an IoT Sensor Model: a network of sensors used in agriculture to check on temperature, soil moisture, and mineral levels. These sensors are connected to a cloud-based server via wireless communication, enabling real-time data transmission and storage. This setup allows continuous monitoring of field conditions, which is essential for precision agriculture. The collected data is processed and analysed in the cloud, where advanced algorithms provide actionable insights for farmers. This cloud-centric approach

facilitates centralized data management, scalability, and accessibility, empowering informed decision-making for optimized irrigation, crop health monitoring, and overall farm management, thus enhancing productivity and sustainability (Sadowski and Spachos, 2020).

Moreover, ground and aerial-based drones have emerged as indispensable tools within agriculture, amplifying a spectrum of agricultural practices encompassing crop health assessment, irrigation, monitoring, spraying, planting, harvest operations, soil, and field analysis. Smart greenhouses, augmented through IoT frameworks, exemplify intelligent climate monitoring and regulation, obviating manual intervention (Eli-Chukwu, 2019). Predictive analytics, integral to intelligent farming, aid farmers in strategic decision-making about crop production, storage, optimal marketing techniques, and risk mitigation. Integrating sensor-derived data, encompassing soil parameters, temperature, pressure, rainfall, and humidity, via dashboards or bespoke mobile applications furnishes farmers with precise soil insights (Ramesh and Vydeki, 2020; Faisal et al., 2023). A recent case study on implementing an AI-based pest management system at a large-scale agricultural farm in California underscores the transformative potential of such technology. Before this intervention, the farm experienced significant crop yield reductions due to persistent pest infestations. A substantial change in their pest management strategy was introducing an AI-driven solution by an agricultural technology startup. This advanced system leveraged machine learning algorithms alongside real-time data collection from sensors and drones to accurately identify and predict pest outbreaks. Consequently, the farm achieved a 40% reduction in pesticide usage, improved overall crop health, and realized a 25% increase in productivity within a year. This case exemplifies AI's profound impact on pest management practices, promoting greater sustainability and efficiency in modern agriculture (Victor et al., 2024).

Artificial Intelligence heralds a multitude of avenues for augmenting farmer incomes and bolstering agricultural productivity, notably by leveraging diverse AI frameworks. Determining soil types and nutrition, crucial crop selection, and quality determinants can be systematically facilitated through diverse AI frameworks. To safeguard cultivated crops from threats posed by animals and intruders, AI-driven surveillance systems adeptly monitor real-time video feeds from agricultural fields, promptly identifying breaches and issuing timely alerts. AI also uses real-time sensor data and visual analytics enabled by drones to predict and find pest infestations before they happen. It also programs autonomous robots to deliver fertilizer precisely, which saves money and increases crop yields. Accurate crop price forecasting based on yield projections holds immense potential in devising cogent pricing strategies and minimizing waste, empowering agricultural stakeholders to negotiate optimal harvest prices. AI's pervasive role across these domains fortifies "Agriculture as a Service", empowering farmers to automate farming practices, adopt precision cultivation techniques, and garner higher crop yields while optimizing resource utilization (Liakos et al., 2018). Table 1 shows the multifaceted utilities of AI delineated across eight principal domains.

Connecting these AI applications to the evolving agricultural landscape dynamics to discern demand and supply nuances The burgeoning demand for data-driven farming and predictive analytics, especially in the post-pandemic landscape, mandates a proactive adoption of precision farming methodologies buoyed by AI and machine learning frameworks. These frameworks underpin comprehensive crop and livestock management, mitigating conjecture through yield forecasting, adept supply chain management, and risk assessment (Chandra and Albaaji, 2023). This knowledge-driven strategy is the perfect example of the paradigm shift that automated analytics has enabled. Blockchain technology, characterized by its decentralized and immutable ledger system, is poised to revolutionize agricultural supply chains by ensuring digital assets' transparent and traceable provenance. Applying blockchain within agriculture facilitates seamless traceability, from seed identification to harvest (Kim et al., 2020). Consumers, increasingly concerned about food quality and sourcing, stand to benefit from blockchain-enabled access to comprehensive crop details, including seed origins, planting specifics, fertilization, harvest timelines, and export details, culminating in QR code-enabled information at the point of sale, offering a transparent end-to-end view of the supply chain.

Leveraging plant growth images and videos is a cornerstone for deploying intelligent models underpinned by machine learning algorithms trained on vast datasets to facilitate real-time predictions. This ingrained intelligence finds widespread application across diverse agricultural activities, encompassing drone and satellite imagery analysis, disease and pest detection, weather forecasting, and optimal timing for fertilizer application. A specialized facet of AI is tasked with predicting optimal chemical spray usage and addressing variations in harvest times and product yields, even under uniform fertilization conditions. The intricate complexities of amalgamated farming issues find resolution through developing and utilizing specialized built-in intelligence models, leveraging in-depth agricultural data, growth patterns, and disease prevalence, culminating in actionable market and harvest forecasts (Furstenau et al., 2023).

2.1. Advanced vs. Traditional farming methods

Advanced agricultural technologies significantly enhance farming efficiency and sustainability compared to traditional methods. Precision farming utilizes GPS-guided machinery and data analytics to optimize inputs like water and fertilizers based on specific field conditions, leading to higher crop yields and more efficient resource use. This approach contrasts with traditional farming's broad and often inefficient input application, which relies heavily on manual labour (Kırkaya, 2020). Intelligent sensors and IoT devices provide real-time data on soil moisture, temperature, and crop health, enabling timely decision-making and adjustments that prevent crop diseases and optimize irrigation. Unlike traditional periodic manual inspections, these technologies offer continuous monitoring, improving resource efficiency and crop management (Sreekantha and Kavya, 2017).

AI and machine learning further revolutionize agriculture by analysing vast data sets to offer predictive analytics, crop modelling, and decision support. These technologies help farmers optimize planting schedules, predict yields, and manage risks more effectively than traditional decision-making processes, which rely on experience and intuition (Virnodkar et al., 2020). By comparing these advanced technologies with traditional methods, it is evident that precision farming, smart sensors, IoT, and AI collectively enhance agricultural sustainability and productivity through targeted, efficient, and data-driven resource management.

2.2. Challenges in adopting technologies in agriculture

Challenges loom large within the ambit of futuristic agricultural sectors, notably the concerted thrust across the supply chain to amplify capacity and slash costs, which might undercut the farmers. A rural farming community in Sub-Saharan Africa exemplifies this. Farmers endeavoured to integrate IoT sensors for precision agriculture in this context but were confronted with severe challenges, such as inadequate internet connectivity and restricted access to data networks. The insufficiency of reliable connectivity severely hindered their capacity to transmit real-time data and access crucial information necessary for informed decision-making. Because of this, using IoT sensors did not have the desired effects, showing how vital robust internet infrastructure is for making advanced agricultural technologies work well (Ahmad and Sharma, 2023). The integration of built-in intelligent technologies, while carrying unique advantages, grapples with several hurdles in its adoption within the agricultural domain, particularly in ASEAN countries, as shown in Table 2.

Additionally, the landscape of AI-based research in Southeast Asian countries presents its own set of challenges. Even in the most developed laboratories, a small community of 50 to 75 professionals operates

Table 1
The multifaceted utilities of AI across eight principal domains.

Domain	Description
1. IoT-driven growth for crop nourishment	AI analyses data from IoT sensors to optimize crop nourishment, offering insights into soil conditions, nutrient levels, and water needs. This helps farmers make data-driven decisions on fertilization and irrigation to boost crop growth and yield.
2. Image-based analysis for monitoring crop health	AI algorithms process images from drones or sensors to assess crop health, detecting diseases, pests, and nutrient deficiencies. This early detection enables timely corrective actions to safeguard and enhance crop health.
3. Formulation of tailored agricultural product blends for superior fertilization	AI customizes fertilizer blends based on soil composition, crop type, and growth stage, ensuring optimal nutrient delivery for healthy growth and maximum yield.
4. Augmented crop health monitoring for maximal yield	AI-powered systems continuously monitor crop indicators like growth rates, leaf colour, and moisture levels, enabling farmers to adjust practices to maximize yield potential.
5. Empowering farmers to optimize irrigation for efficient resource utilization	AI analyses soil moisture, weather forecasts, and crop water needs to optimize irrigation schedules, conserve water, reduce costs, and ensure optimal crop hydration.
6. Implementation of autonomous technological interventions in agriculture	AI-driven technologies, such as robotic systems and automated machinery, enhance planting, harvesting, and spraying efficiency, reducing labour costs and improving farm operations.
7. Strategic direction for value-added products, manufacturing, and enterprise marketing	AI aids in developing value-added products, optimizing manufacturing, and implementing marketing strategies by analysing market trends and consumer preferences, helping farmers diversify offerings and increase profitability.
8. Perceptive comprehension of the market	AI analyses market data, consumer behaviour, and supply chain information to provide insights into demand fluctuations and supply chain dynamics, guiding production planning, pricing, and market positioning.

Table 2
Challenges in integrating advanced technologies into agricultural practices.

Challenge	Description
1. Connectedness and data availability	The fundamental challenge is ensuring robust connectivity and data availability across agricultural setups.
2. Financial stability	The financial underpinning for integrating advanced technologies remains a hurdle, especially for smaller farming communities.
3. Reliability of systems and technology	Ensuring the dependability and resilience of integrated systems and technology poses a significant challenge.
4. Security and acceptance of information	Concerns about the security of data and the broader acceptance of information among stakeholders hinder seamless adoption.
5. Data Privacy, Storage, and Use	The ethical and regulatory aspects of data privacy, secure storage, and appropriate utilization present formidable challenges.
6. Lack of technical expertise	A shortage of technical know-how and expertise impedes the effective implementation of sophisticated technologies.
7. Social acceptance and recognition	Garnering social acceptance and recognition for these advanced systems remains challenging.
8. Cost-Effectiveness	Ensuring the cost-effectiveness of integrating these technologies into agricultural practices is a significant concern.
9. Ease of use and training	Ensuring ease of use and providing adequate training for stakeholders remain essential for successful adoption.
10. Unethical Stakeholders	Mitigating the impact of unethical stakeholders poses a significant challenge in integrating these technologies.
11. The unsuitable facilities	The lack of infrastructure limits modern farming technology. Without solid infrastructure, these technologies cannot maximize their benefits, resulting in inefficiencies and poor agricultural yield and sustainability.

within this domain, hindering significant advancements. Factors such as limited space, resource scarcity, administrative barriers, research methodologies, subpar computing facilities, inadequate quality data interpretation, and insufficient fund allocation in national budgets underscore the predicaments afflicting Artificial Intelligence research in agriculture. Translating these technologies from the laboratory environment to real-world scenarios is imperative to propagate the growth of Artificial Intelligence applications in agriculture, ensuring that all stakeholders derive benefits. The deployment of recommendation systems benefits consumers and farmers alike, streamlining the entire supply chain swiftly and efficiently. Integrating Artificial Intelligence-driven methodologies across diverse farming stages is poised to usher in an era of flourishing agricultural resources, necessitating the simultaneous advancement of multiple research methodologies. Furthermore, the escalating importance of agriculture and the surge in demand for food grains in the post-pandemic era underline the need for novel methods and AI-based frameworks in agriculture. We must push for new technological frameworks and improvements in farming methods and marketing strategies to get around problems like limited transportation, scarce raw materials, and trouble deploying skills. This will help the national economy grow.

2.3. Artificial intelligence-based framework

A 2020 economic survey that emphasized the creation of farming models based on sizeable datasets highlighted the necessity of data-based farming to achieve high yields (Gaitan et al., 2015). Integrating Artificial Intelligence into agriculture and the agricultural market presents potential advancement. This implementation has already found practical application in various developed countries (Javaid et al., 2023).

Fig. 3 depicts a model framework called Systems of Sedentary behaviours (SoS), illustrating rainfall prediction in specific areas using Artificial Intelligence techniques like artificial neural networks. Such methodologies, developed by meteorological departments, contribute significantly to rainfall and climate change forecasting. When incorporated into the SoS framework, these technologies yield maximal agricultural profits and higher crop yields for farmers (Priyadarshini and Abhilash, 2020), imparting a sense of direction to the forthcoming generations.

Artificial Intelligence in agriculture encapsulates various subfields such as natural language processing, robotics, machine learning, automatic reasoning, knowledge representation, expert systems, computer vision, speech recognition, automated data analytics, cloud computing,

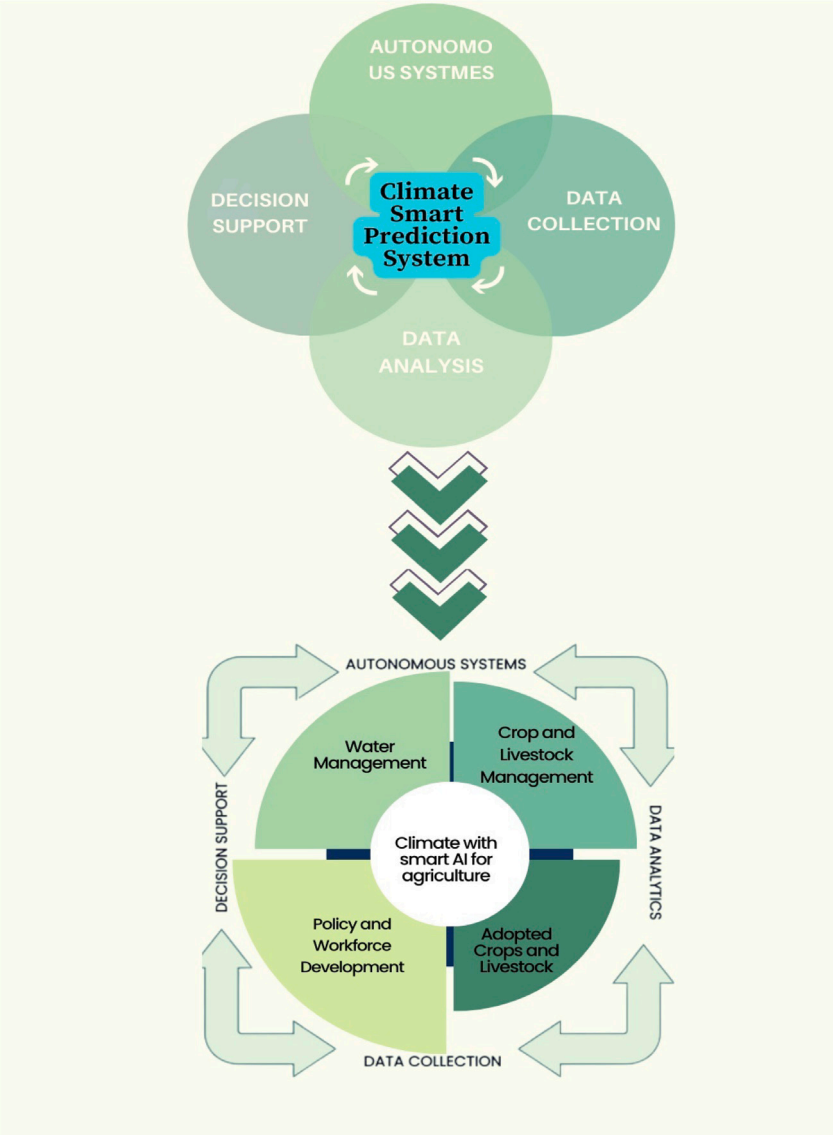


Fig. 3. Artificial intelligence-based SoS.

statistical computing, and deep learning (Gaitan et al., 2015). The technological software that farmers use integrates many of these subfields. For instance, standalone Q&A systems in diverse languages empower farmers. At the same time, deep learning-based agriculture mapping from satellite images provides intelligent forecasting systems for soil science, crop yield, pest management, and harvesting.

The SoS framework relies fundamentally on data and analytics. It incorporates diverse environmental factors—dew, sunlight, rainfall, and other climatic changes—analysed via sensors and satellite images. The subsequent data processing by an analytical engine, which causes trained models to start particular tasks, emphasizes how dependent prediction and automation are on data availability. The knowledge acquired is instrumental in developing decision support systems benefiting agriculture and allied services like water and livestock management.

Precision agriculture emphasizes enhancing overall harvest quality and accuracy through AI methods. AI can be used in agriculture to find weeds with sensors connected to herbicides and make intelligent predictions about where to store seeds and how to distribute grains (Mahmud et al., 2023). Crop diseases pose a severe challenge, leading to diminished productivity. Machine learning techniques facilitate early identification and remedial solutions for crop diseases, ensuring higher yields (Gill et al., 2017). Furthermore, farmers' dependency on external factors and the perishable nature of agro-products necessitate AI-driven solutions for optimal storage, transportation, and demand forecasting. Predicting market needs during specific events or festivals can also aid in foreseeing demand–supply dynamics, thereby creating new market opportunities (Lin et al., 2017).

A case study of a precision irrigation system in a Napa Valley vineyard demonstrates the substantial benefits of advanced agricultural technology. The vineyard implemented a Vititech Solutions system that uses real-time data from soil moisture sensors and weather forecasts to deal with water scarcity and inconsistent grape quality. This AI-driven system optimizes water distribution, ensuring each vine receives the necessary amount. Consequently, the vineyard achieved a 30% reduction in water usage and a 20% improvement in grape quality within two seasons. This case underscores the potential of precision irrigation to enhance sustainability and productivity in viticulture (Seyar and Ahamed, 2023). Thus, the application of technology, particularly AI, holds promise for farmers with limited landholdings, aiding in weather prediction at the local level and guiding sustainable farming practices. This technological integration assists in pest management, robotics for multi-crop harvesting, demand prediction, and meeting regional and export requirements.

Table 3
Southeast Asian global rankings for connectedness and data flows.

Country	Connectedness index	Dataflow	GDP/Capita rank	Reference
Singapore	High	High	1	world bank (2023)
Malaysia	Upper-middle	Medium	61	ITU (2023)
Thailand	Upper-middle	Medium	93	world bank (2023)
Vietnam	Lower-middle	Low	126	world bank (2023)
Indonesia	Lower-middle	Low	170	WDI (2023)
Philippines	Lower-middle	Low	132	world bank (2023)

3. Agricultural intelligence model

This study is centred on Southeast Asian countries that rely heavily on agriculture for economic stability. Comprising eleven nations—Brunei, Burma, Cambodia, Timor-Leste, Indonesia, Laos, Malaysia, Philippines, Singapore, Thailand, and Vietnam—this region reflects remarkable diversity in culture, technology, farmland, and economy. Notably, Malaysia, Philippines, Singapore, Thailand, Indonesia, and Vietnam contribute over 70% of their income through agri-food, showcasing their technological vibrancy and extensive farmland. The focal point of this manuscript on these six countries stems from their economic stability, technological adaptability, and ample farmland availability for multiple crops. Their remarkable economic upsurge substantiates these nations' significant emphasis on their agricultural sectors in recent years. Critical to these countries is ensuring food security. In 2019, the agri-food sector contributed a staggering $717 billion to the collective economies of Indonesia, Thailand, the Philippines, and Vietnam, marking a 30% increase from 2015. Moreover, this sector employed 48% of the total workforce across these countries, generating 127 million jobs. The Association of Southeast Asian Nations (ASEAN) emerges as a globally productive agricultural region, and integrating technological solutions will catalyze exponential market growth.

Connectivity and data flow are instrumental in fostering growth and enabling an agricultural boom in these nations. Fortunately, prominent players in this sector rank within the top 50 regarding connectedness, indicating a promising landscape if these countries receive adequate technological training. The connectedness index of ASEAN countries is depicted in Table 3 (Source: International Monetary Fund, World Economic Outlook Database).

"New standards for new-age farming" epitomizes Agriculture Revolution 5.0. The diverse standalone technologies mentioned serve as crucial aids for farmers. Over 40 critical farming activities require analysis before making decisions, ensuring optimal yields and returns on investment (Chohan et al., 2020). Farmers' obstacles include difficulties such as low-quality inputs, the unpredictability of the weather, incorrect forecasting rules for crucial phases of crop development, and intermediate problems such as a lack of available labour and inadequate insurance coverage. Efficient access to real-time data encompassing weather, crop characteristics, soil, and pest data, coupled with new and emerging technologies like drones, cloud computing, and image analytics, has transformative potential for traditional agriculture. This convergence of technology and new-age products could redefine traditional agriculture into data-driven precision farming, ensuring profitability and sustainability.

Digital agriculture is a panacea for the multifaceted problems haunting the agricultural domain. Digitization promises farmers enhanced access to markets, inputs, advisory services, and credit. However, the lack of timely and accurate data remains a bottleneck, hindering the establishment of a demand-driven, efficient supply chain. Mitigating these challenges necessitates synthesizing data for informed decision-making by various stakeholders, facilitated by digital technologies and numerous AgriTech startups. Addressing pest control remains a significant challenge in this sector. Although not entirely automated, technology, notably AI facilitated through drones, can substantially enhance pest control efficiency, ensuring safer, quicker, and cost-efficient solutions for farmers (Li et al., 2020). AI-powered deep learning algorithms combined with digital image processing help identify diseases and find the right treatments (Subeesh and Mehta, 2021). This lets farmers find problems quickly and fix them, preventing possible epidemics.

4. Results

The paradigm of modern agriculture is evolving by integrating cutting-edge technologies into farm operations. The proposed AI-based model for agriculture, which shows the stages of its implementation in Fig. 4, incorporates advanced technologies into a cohesive system designed to enhance productivity and sustainability. Initially, data is collected through sensors and IoT devices, capturing real-time information on soil moisture, temperature, humidity, pest activity, and crop health. Fig. 5 represents samples of sensors used in this stage. This data is then transmitted and stored using edge computing for immediate processing and cloud storage for centralized management. Subsequently, machine learning algorithms and big data analytics analyse the data, uncovering patterns and generating actionable insights. These insights feed into a decision support system (DSS), which provides AI-driven recommendations through a user-friendly interface, facilitating informed decision-making for farmers. Automated systems and alert mechanisms are then activated to execute necessary actions, such as adjusting irrigation or deploying pest control measures. The model incorporates a feedback loop that ensures continuous learning and improvement by integrating outcome data and farmer input, thus adapting to evolving conditions and practical insights.

Fig. 6 shows an advanced agriculturally intelligent farm that uses an Application Programming Interface (API) and a network of technological parts, including a crop-connected sensor network, to run the farm. This integration is the backbone for comprehensive data analysis, empowering the farm's operations. The orchestration of multiple sensors, activated by diverse alarms, is a cornerstone of ensuring the operational efficiency of this system; Fig. 5 represents samples of sensors that were used.

Notably, these sensors play distinct roles, with soil moisture sensors triggering precise watering mechanisms and image analysis to evaluate crop health and instigating interventions such as medication, pest control, and fertilizer administration. Furthermore, the adaptive regulation of the greenhouse, responding to UV exposure by opening or closing as needed, is a testament to the system's dynamic responsiveness. The meticulous recording of each analysis into a blockchain repository, made possible by node entry, is a crucial component of this agricultural framework. This approach meticulously traces the entire life cycle of crops, commencing from seed selection to harvest and eventual retail. Subsequently, these meticulously documented data traverse to cloud analytics, presenting predictions readily accessible to farmers through web-based platforms or applications installed on their smart devices. The accuracy and reliability of these predictions are contingent upon the robustness of the sensor networks and the efficacy of the data transmission frameworks employed. Delineating the sensors essential to intelligent farming reveals a multifaceted toolkit integral to farm management. Acoustic sensors have emerged as versatile tools, contributing significantly to soil cultivation, weeding, and fruit harvesting. Their adaptability and swift response rates, mainly when employed with portable equipment, render them cost-effective solutions across various farm management facets. Adding ultrasonic ranging sensors and a camera for finding weeds expands the range of functions even more. This lets you keep track of tank levels, evaluate crop canopy, and measure spray distance. Also, adding airflow sensors helps figure out things about the soil, like how absorbent it is to air, how much water is in it, and its structure, which makes it possible to tell the difference between different soil types. Elective sensors can record electrical conductivity and transient electromagnetic responses very well. This makes variable-rate applications possible, which is essential in situational farming situations. Additionally, deploying Eddy covariance-based sensors quantifies gas and energy exchanges between

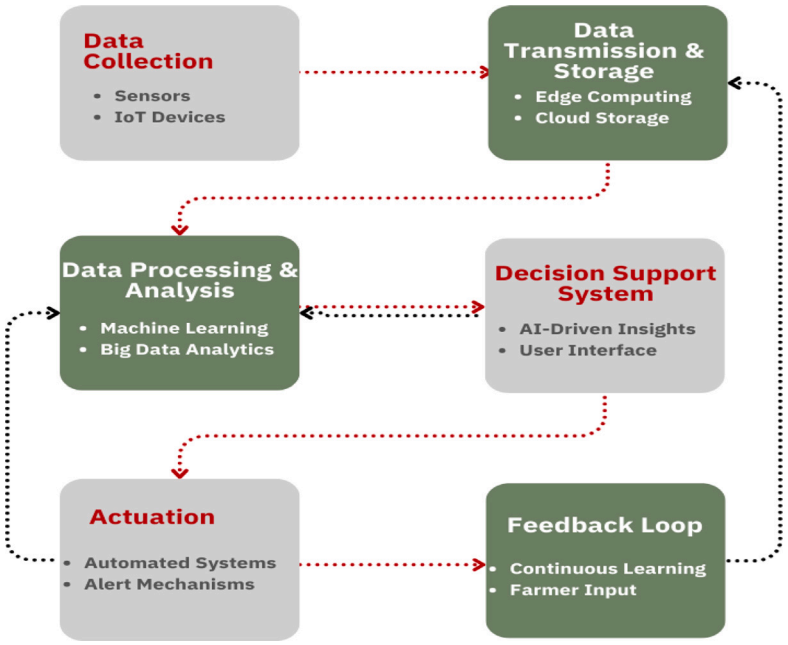


Fig. 4. The fundamental phases of agricultural intelligence model.

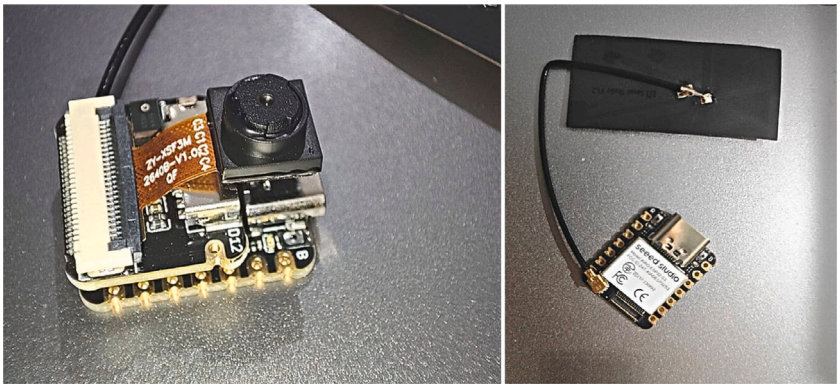


Fig. 5. Examples of sensors specifically designed for data collection in precision farming.

the Earth's surface and the atmosphere, adding depth to environmental monitoring within agricultural contexts.

The contributions of artificial intelligence technologies to the agricultural sector appear in many aspects, incorporating data such as a 30% reduction in water usage and a 20% improvement in grape quality due to AI-driven precision irrigation, as well as the $717 billion economic contribution of the agri-food sector in Southeast Asia, which has increased by 30% since 2015 and created 127 million jobs, would significantly enhance the validity and impact of the findings, furthermore the pervasive utilization of Light Detection and Ranging (LIDAR) technology across agricultural applications is notable, encompassing activities such as land mapping, determination of soil types, 3D farm modelling, erosion monitoring, and yield forecasting. Similarly, remote sensing sensors are essential in storing and analysing geographic data and find extensive application in crop assessment, yield modelling, and forecasting endeavours (Weiss et al., 2020).

4.1. Discussion

In this study, the advanced agriculturally intelligent farm operates through a sophisticated network of technologies, primarily a crop-connected sensor network linked to the farm's API. This integration is the cornerstone for comprehensive data analysis and efficient farm operations. Diverse sensors, activated by specific alarms, play crucial roles in optimizing various aspects of farm management. Soil management sensors encompass a range of functionalities, from enabling precise watering based on soil conditions using moisture sensors to utilizing image analysis for crop health assessment, thereby triggering interventions like medication, pest control, and fertilizer administration. Additionally, the adaptive response of greenhouse sensors to UV exposure dynamically optimizes growing conditions, showcasing the system's responsiveness to environmental cues. The meticulous recording of analyses in a blockchain repository ensures secure and traceable records of the crop's entire life cycle. At the same time, data transmission to cloud analytics facilitates the delivery of accurate predictions accessible to farmers through web pages or smart device applications. The accuracy of these predictions relies significantly on robust sensor networks and efficient data transmission frameworks. An array of essential sensors in intelligent farming includes acoustic sensors with versatile functionalities in soil cultivation, weeding, and fruit harvesting. Their cost-effectiveness and swift response rates, especially with portable equipment, render them valuable for diverse farm management tasks. Further integration of cameras and ultrasonic sensors aids in weed detection, tank level tracking, crop canopy assessment, and spray distance measurement. Moreover, airflow sensors ascertain critical soil characteristics, such as air permeability, moisture percentage, and structure, enabling the differentiation between

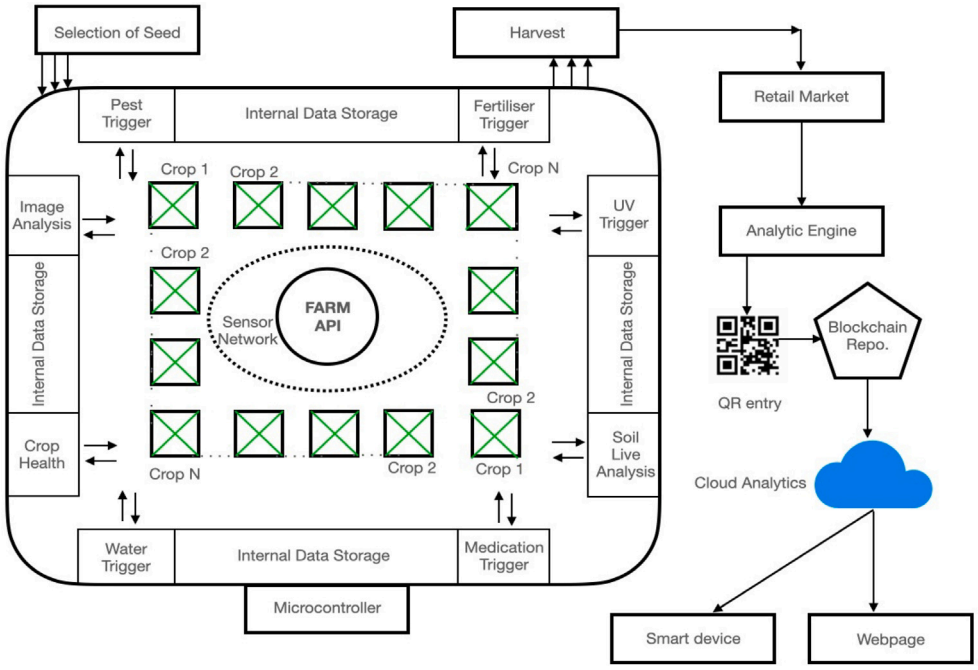


Fig. 6. Agricultural intelligence prototype.

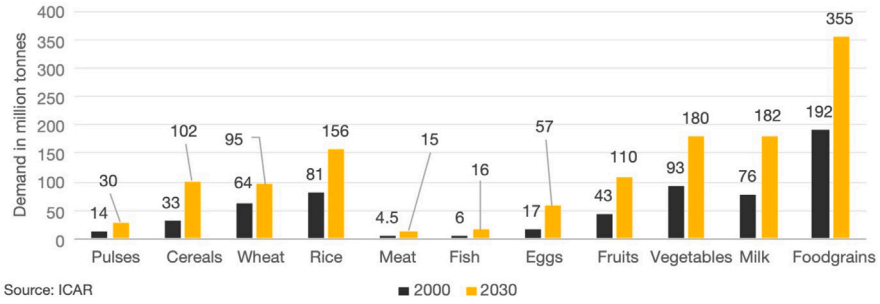


Fig. 7. Predictive demand.

soil types. Electromagnetic sensors record electrical conductivity, facilitating variable-rate applications in situational farming scenarios. Additionally, eddy covariance-based sensors quantify gas and energy exchanges between the Earth's surface and the atmosphere, offering insights into environmental dynamics. The extensive application of LIDAR technology in agricultural domains spans activities such as land mapping, 3D farm modelling, erosion monitoring, and yield forecasting. Likewise, remote sensing sensors are pivotal in storing and analysing geographic data supporting crop assessment, yield modelling, and forecasting. Looking ahead, the emergence of Agriculture 5.0 emphasizes the necessity for self-sustaining autonomous farms, promoting ease, viability, and heightened productivity. Given that the demand for food grains is expected to rise by 2050, almost doubling the current production, it becomes more apparent that we need to switch to self-sufficient digital farming methods. Fig. 7 shows the predicted demand for 2050 in metric tons, emphasizing the urgent need to take proactive steps to transform agriculture. This transition is crucial to addressing the anticipated agricultural challenges in the coming decades.

5. Conclusion

The global landscape is witnessing a concerted effort towards ushering in next-generation agricultural strategies, prominently featuring the deployment of artificial intelligence-based frameworks. To optimize agricultural yield, a crucial recommendation surfaces: an escalated impetus on bolstering research and development endeavours across allied domains of agriculture. Concurrently, enhancing interaction among stakeholders at multiple supply chain echelons is pivotal to benefiting farmers comprehensively. This envisioned work espouses the imperative of a self-sustaining ecosystem and introduces a comprehensive framework model underpinning agricultural intelligence. An automated farmland system stands out as the linchpin for ensuring profitability and bolstering productivity within the agricultural domain. Adopting a Systems of Sedentary Behaviours (SoS) framework is a cornerstone for establishing a robust and refined system architecture. At the crux of this paradigm shift lies the imperative to comprehend innovative and intelligent processes from diverse agricultural universities and research centres and discern their tangible impact on agricultural practices. However, amid the integration of these sophisticated technological paradigms, the challenge persists in empowering farmers and cultivating their awareness regarding these transformative technologies. A concerted, collective effort emerges as the linchpin in addressing this challenge and harnessing the dividends of technological advancements for the agricultural community. As an unintended but significant side effect, the suggested method pushes for a more complete view that includes agro-ecology, agro-forestry, and expanding the range of livestock markets. Notably, the roadmap to realizing Agriculture 5.0 hinges

on the collective adoption of an integrated, multi-dimensional research approach spanning agriculture and its allied realms. This holistic amalgamation of advanced technologies and progressive methodologies sets the stage for a promising and vibrant Agriculture 5.0 era.

CRediT authorship contribution statement

Vinod Chandra S.S.: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Anand Hareendran S.:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ghassan Faisal Albaaji:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Ethics approval

This manuscript does not contain any studies with animals, and no harm has been done during any phase of the work.

Consent for publication

All authors have provided written consent for publishing this manuscript in this journal.

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